ALICE L WALTON SCHOOL OF MEDICINE

Topoisomerase Inhibitors, Microtubule Inhibitors, and Antitumor Antibiotics

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Adapted by: Diptiman Bose, Trager Hintze, Chelsey Deel

Editors: Jordan Finn, Russell Johnson

Adapted from: Topoisomerase Inhibitors, Microtubule Inhibitors, and Antitumor Antibiotics

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Learning Objectives (2) Versions

After completing this brick, you will be able to:

1. Explain the mechanism of action of agents of topoisomerase inhibitors, microtubule inhibitors, and antitumor antibiotics.
2. Explain the adverse effects of topoisomerase inhibitors, microtubule inhibitors, a

CASE CONNECTION

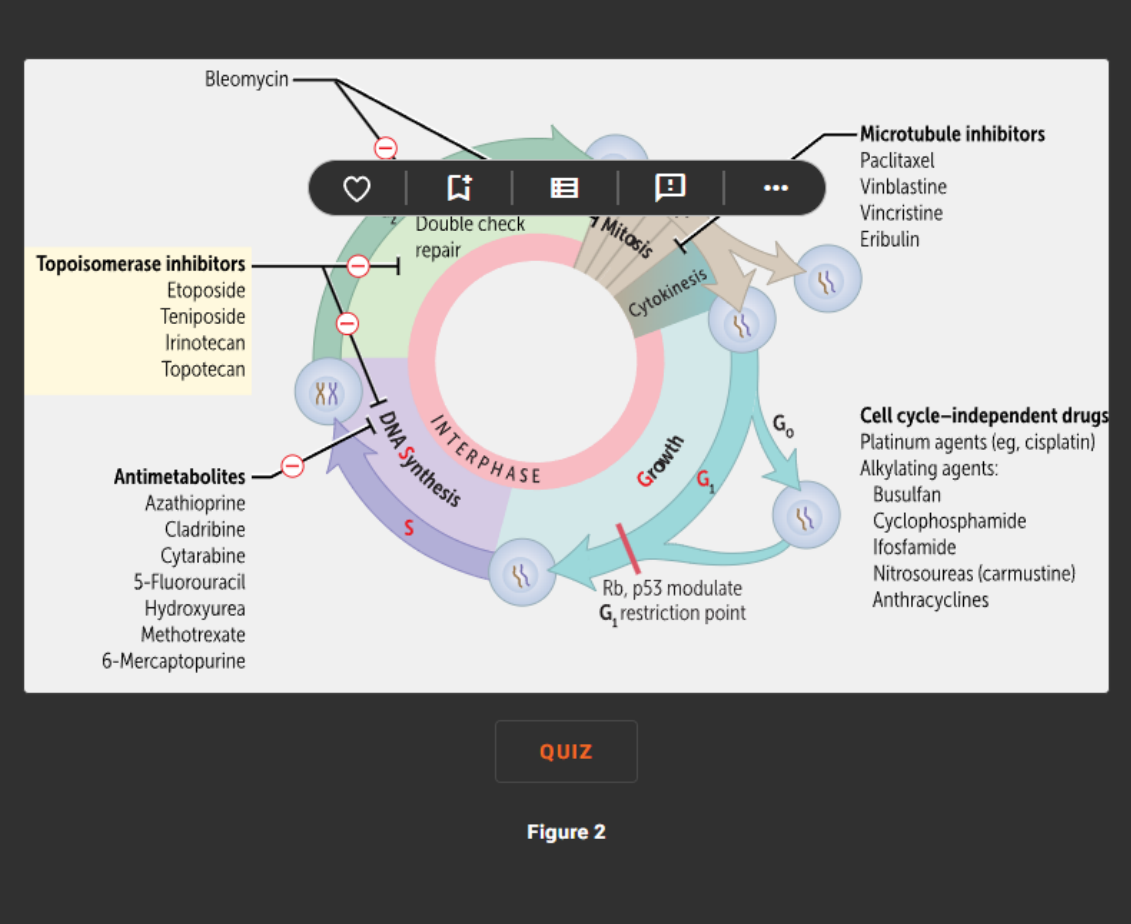
BV, a 60-year-old man is diagnosed with non-Hodgkin's lymphoma. The Tumor Board decided that bleomycin should be added to BV's treatment regimen. As a 4th-year medical student, you accompanied the BV's oncologist to discuss his treatment. BV was anxious about this outcome and treatment. He was upset because he had read how these treatments can make you lose weight, hair, and have other long-lasting effects. The oncologist assures BV that his team will be there for him and will do their best to take care of him. The oncologist asks you to enter an order for a pulmonary function test for BV before treatment. BV says, "I know about the blood work, but why a lung test?! What is going on?"

What is the reason for ordering a lung function test for BV? How would you reassure BV?

Consider your answer as you read, and we'll revisit at the end of the brick.

GO TO CONCLUSION ↓

We are going to discuss three different classes of chemotherapeutic agents. The anthracyclines are cell-cycle non-specific, while topoisomerase inhibitors and microtubule inhibitors are cell-cycle specific.



QUIZ

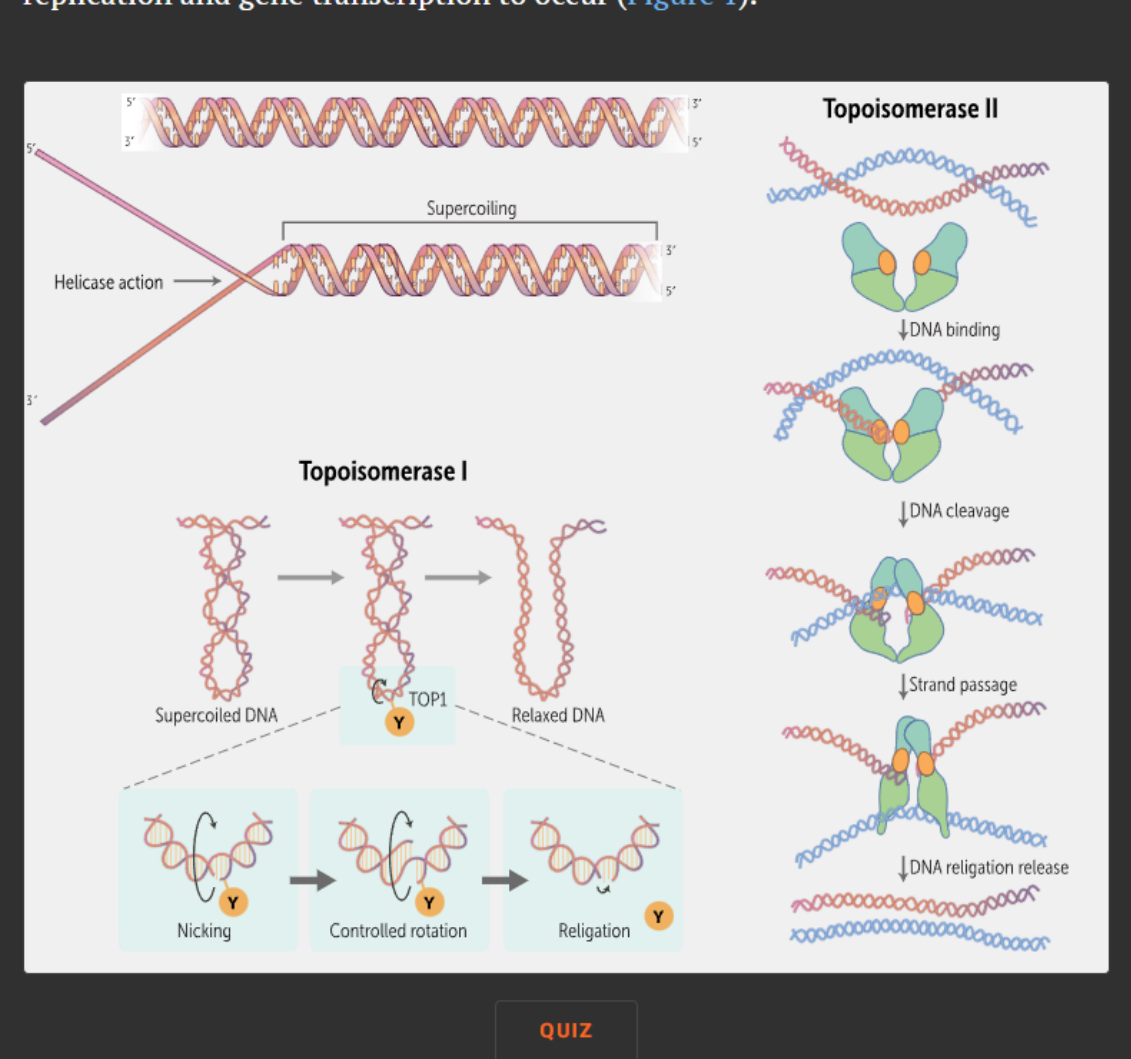
Figure 2

What Are Topoisomerase Inhibitors? (LO#1)

Topoisomerase inhibitors kill cancer cells by interfering with topoisomerases, which are enzymes that block the normal release of tension caused by supercoiling in the DNA double helix. This release of tension is important in DNA replication, transcription, and repair. These drugs are separated into topoisomerase I and topoisomerase II inhibitors, depending on the enzyme they affect.

Mechanism of Action of Topoisomerase Inhibitors

Topoisomerase I clea ix and religates the strand without requiring adenosine triphosphate (ATP) hydrolysis. In contrast, topoisomerase II cleaves both strands of the DNA, religates each strand, and requires ATP. Both enzymes relax the tension on the coiled DNA helix, allowing replication and gene transcription to occur (Figure 1).



QUIZ

Topoisomerase I Inhibitors. Drugs like **topotecan** and **irinotecan** bind to topoisomerase I and prevent the religation step. This leads to a build-up of single-strand breaks. When a DNA replication fork meets one of these breaks, an irreversible double-strand DNA break occurs and causes cell death.

Topoisomerase II Inhibitors. Drugs like **etoposide** and teniposide inhibit religation of the double-strand break created by the enzyme during the S phase or the early G_2 phase of the cell cycle (Figure 2).

Mnemonic

Topoisomerase I causes a ONE-strand break.
Topoisomerase II causes a TWO-strand break.

Adverse Effects of Topoisomerase Inhibitors(LO#2)

As with most chemotherapy drugs, the topoisomerase inhibitors commonly cause **myelosuppres** ow peripheral blood cell counts), nausea, vomiting, and fatigue. **Diarrhea** is a major adverse effect of topoisomerase inhibitors, especially irinotecan (~82% of patients). Early or acute diarrhea occurs shortly after administration and shows cholinergic symptoms such as bradycardia, hypersalivation, miosis, lacrimation, and abdominal cramping. Atropine can be used to treat such symptoms. Late diarrhea can lead to dehydration, electrolyte imbalance, and sepsis. Etoposide can cause secondary leukemia at high doses. The late onset is caused by the accumulation of irinotecan's metabolite (SN-38). This metabolite damages the intestinal mucosal lining causing diarrhea.

What are the main topoisomerase I and II inhibitors?

What Are Microtubule Inhibitors? (LO#1)

Antimicrotubule drugs do exactly as their name suggests—target cellular microtubules. Microt vesicles along the cytoskeleton and to form the mitotic spindle during mitosis, specifically metaphase or M phase. In metaphase, DNA is separated from central alignment to its respective poles by microtubules to eventually form two daughter cells. Antimicrotubule drugs inhibit the action of the microtubules and prevent this step from being completed, which results in cellular arrest in metaphase. This leads to apoptosis (programmed cell death). Two of the subclasses of antimicrotubule drugs, vinca alkaloids and taxanes, primarily target the mitotic spindle.

Mechanism of Action of Microtubule Inhibitors

Microtubules are made up of α- and β-tubulin subunits. These subunits are in dynamic equilibrium, polymerizing and depolymerizing to allow for the cell to complete its tasks (Figure 3). The depolymerization step is key for anaphase and the separation of the chromosome.

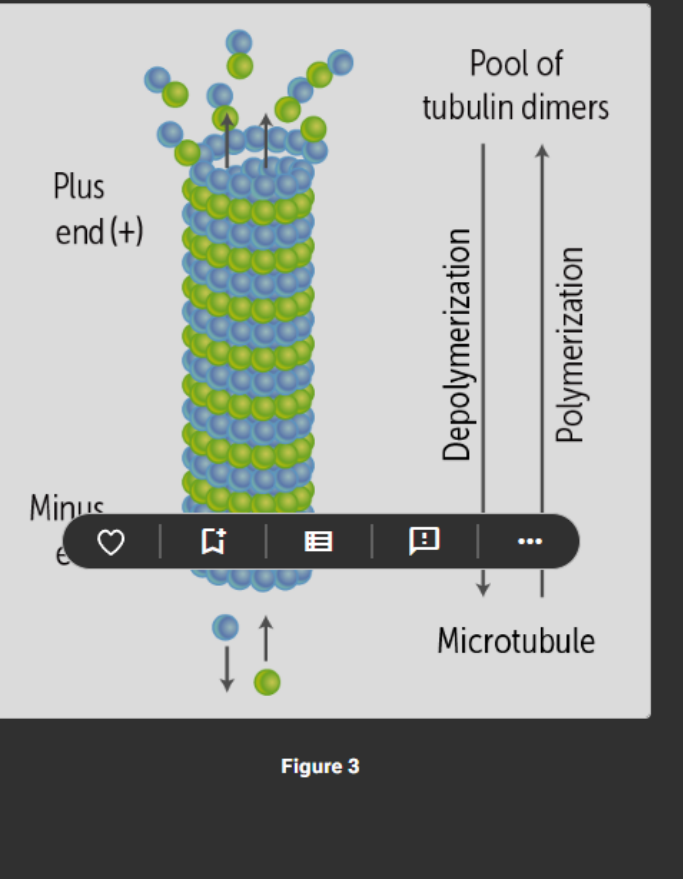


Figure 3

Vinca alkaloids bind to β-tubulin and inhibit its polymerization of tubulin (Figure 4) to form the microtubules. The tubulin dimers aggregate, and the spindle apparatus is frozen in metaphase. The chromosomal segregation and cell proliferation are thus inhibited.

In contrast, **taxanes** bind to β-tubulin and prevent mitotic spindle depolymerization. The inhibition of polymerization and stabilization of the polymer leads to accumulation of the microtubules. Hence, when stabilized by taxanes, the microtubules are nonfunctional, and the chromosomes cannot separate in mitosis, leading to cell death.

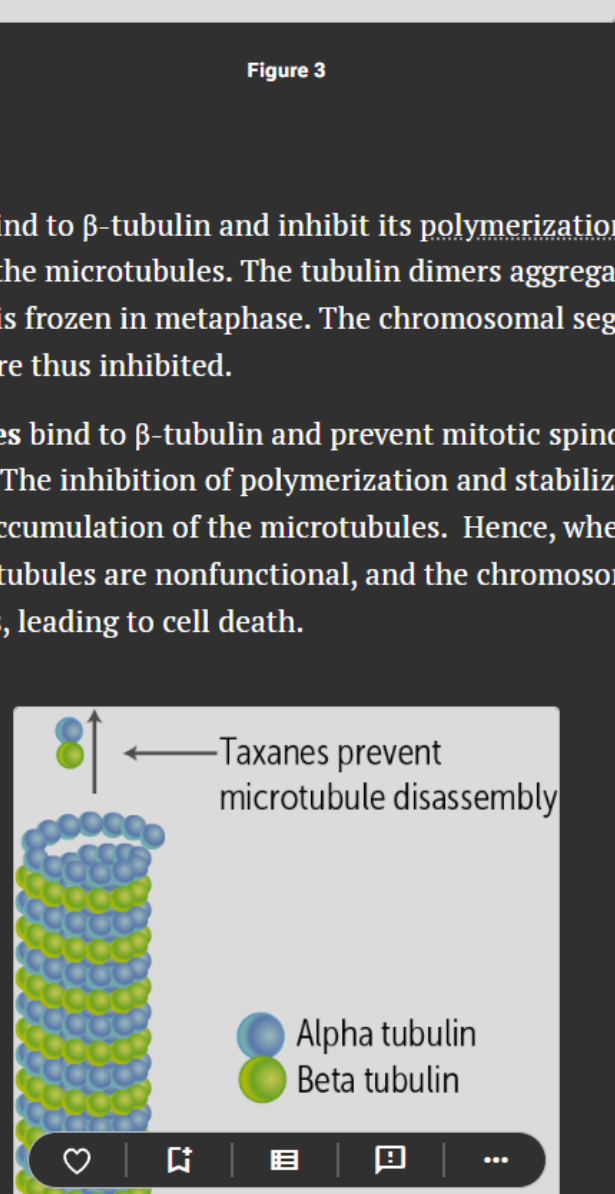
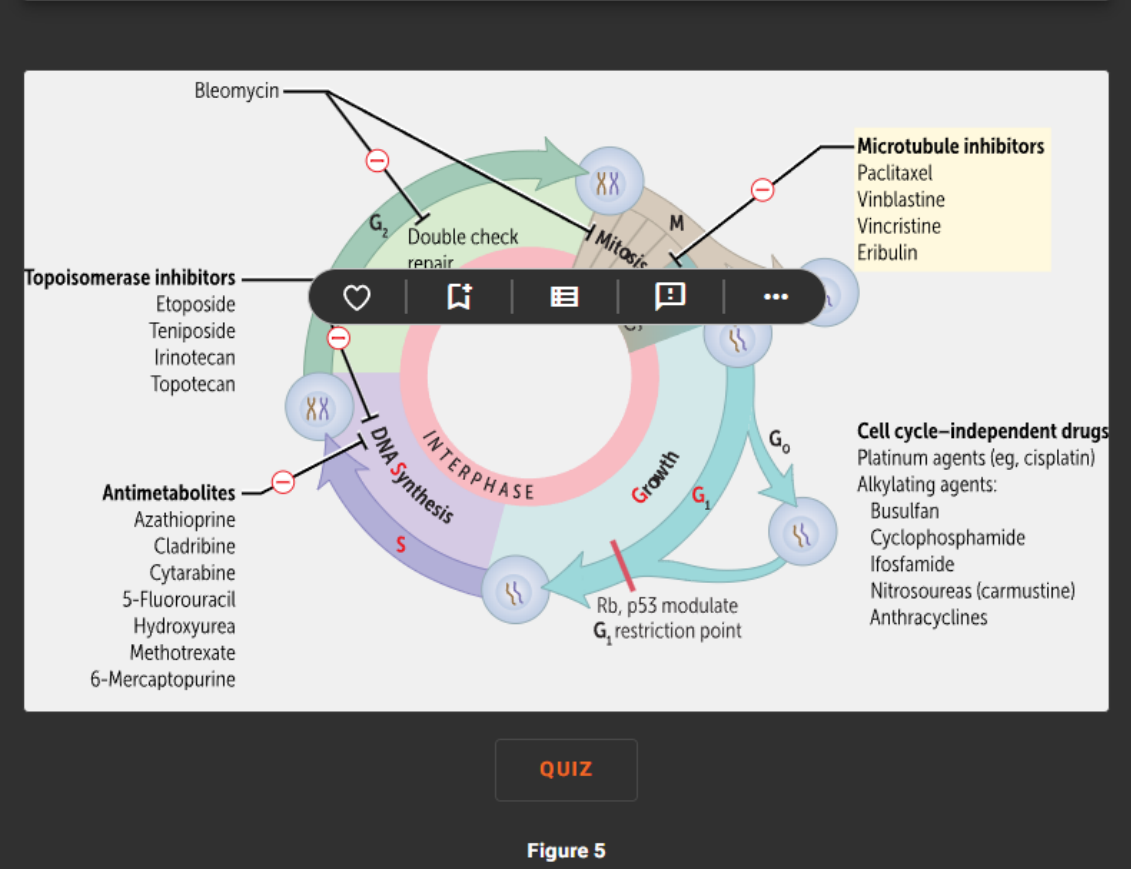

Taxanes prevent microtubule disassembly
Alpha tubulin
Beta tubulin
Vinca alkaloids prevent microtubule assembly

Figure 4

How do vinca alkaloids and taxanes differ in their mechanism of action?



QUIZ

Figure 5

What are some common examples of microtubule inhibitors?

Vinca alkaloids : *Vincristine, Vinblastine, and Eribulin*

Taxanes: *Paclitaxel, Docetaxel, Cabazitaxel.*

Adverse Effects of Microtubule Inhibitors (LO#2)

Vinca alkaloids commonly cause hair loss but can more significantly lead to **peripheral neuropathy** (numbness, tingling, and weakness in the hands and feet) that is usually reversible. In rare cases, patients may develop hyponatremia associated with high levels of antidiuretic hormone (Syndrome of Inappropriate Antidiuretic hormone=SIADH). Constipation is also common with the use of Vincristine.

Paclitaxel and doceta like neutropenia and leukopenia, with **peripheral neuropathy** being a common adverse effect. Due to serious **hypersensitivity reactions**, patients on paclitaxel should be premedicated with dexamethasone, diphenhydramine, and an H2 receptor antagonist. Docetaxel can lead to **severe fluid retention** (peripheral edema or pericardial effusions), so patients should be pretreated with dexamethasone.

What is the most significant adverse effect of vincristine?

What Are Antitumor Antibiotics? (LO#1)

The somewhat oddly named antitumor antibiotics do not target bacterial infections but are effective chemotherapy agents. Many end with -mycin, just like the familiar antibiotic erythromycin (which is not used for cancer, as we know). The most common of these are anthracyclines, among the most effective chemotherapeutic agents in the repository, as well as bleomycin and mitomycin C.

Anthracyclines

Anthracyclines, such as **doxorubicin** and **daunorubicin**, intercalate into double-stranded DNA and inhibit both DNA and RNA synthesis. The anthracyclines are cell cycle–independent agents. This action leads directly to apoptosis. But they also have additional antitumor effects. Anthracyclines can create **free radicals** and cause lipid peroxidation and DNA strand injury.

Adverse Effects of Anthracyclines (LO#2)

Besides the usual adverse effects of chemotherapy drugs (eg, myelosuppression), the main concern with anthracyclines is irreversible, dose-dependent **cardiotoxicity.** The cardiotoxicity include cardiomyopathy, pericarditis, myocarditis and left ventricular heart failure. **Dexrazoxane** is used as a chemoprotective agent used to prevent the doxorubicin-induced cardiotoxicity. It is a chelating agent that interferes with iron-mediated free radical generation, thus protecting the tissue against anthracycline-induced cardiotoxicity.

What are the two main adverse events that occur with anthracycline use?

Mitomycin C

This drug acts like an alkylating agent; it alkylates DNA and produces interstrand cross-linking. Mitomycin C is not a cell-cycle specific agent.

Adverse effects of Mitomycin C

Hemolytic-uremic syndrome which can result in renal failure can occur. Myelosuppression, mucositis, anorexia and fatigue can occur.

CLINICAL CORRELATION

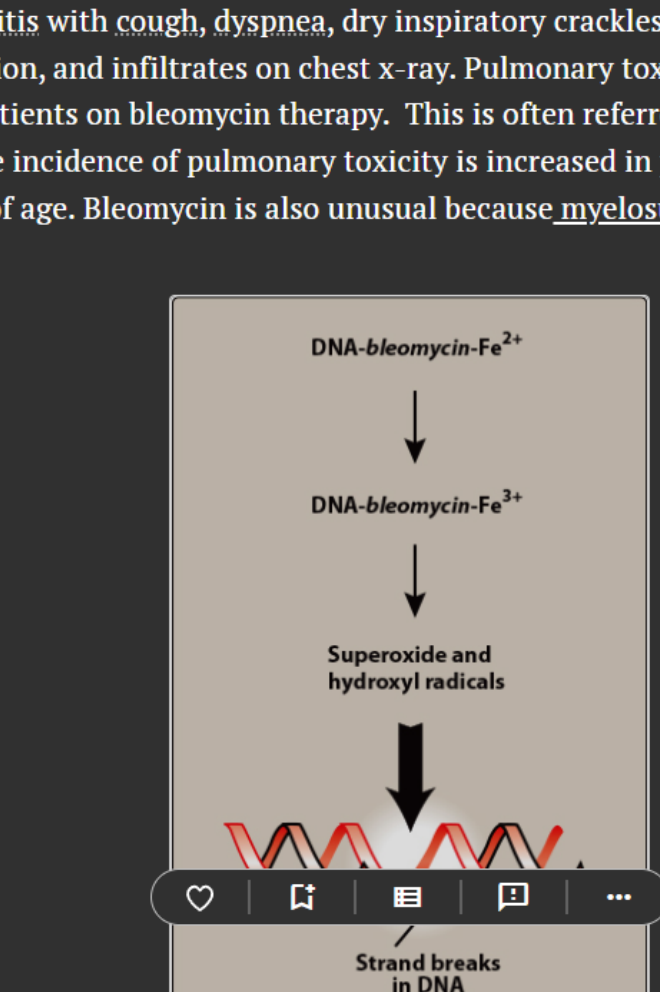
A patient taking mitomycin C who presents with pale conjunctivae, fatigue, petechiae, and oliguria should raise suspicion for hemolytic uremic syndrome, which is a combination of hemolytic anemia, renal failure, and bloody diarrhea.

Bleomycin

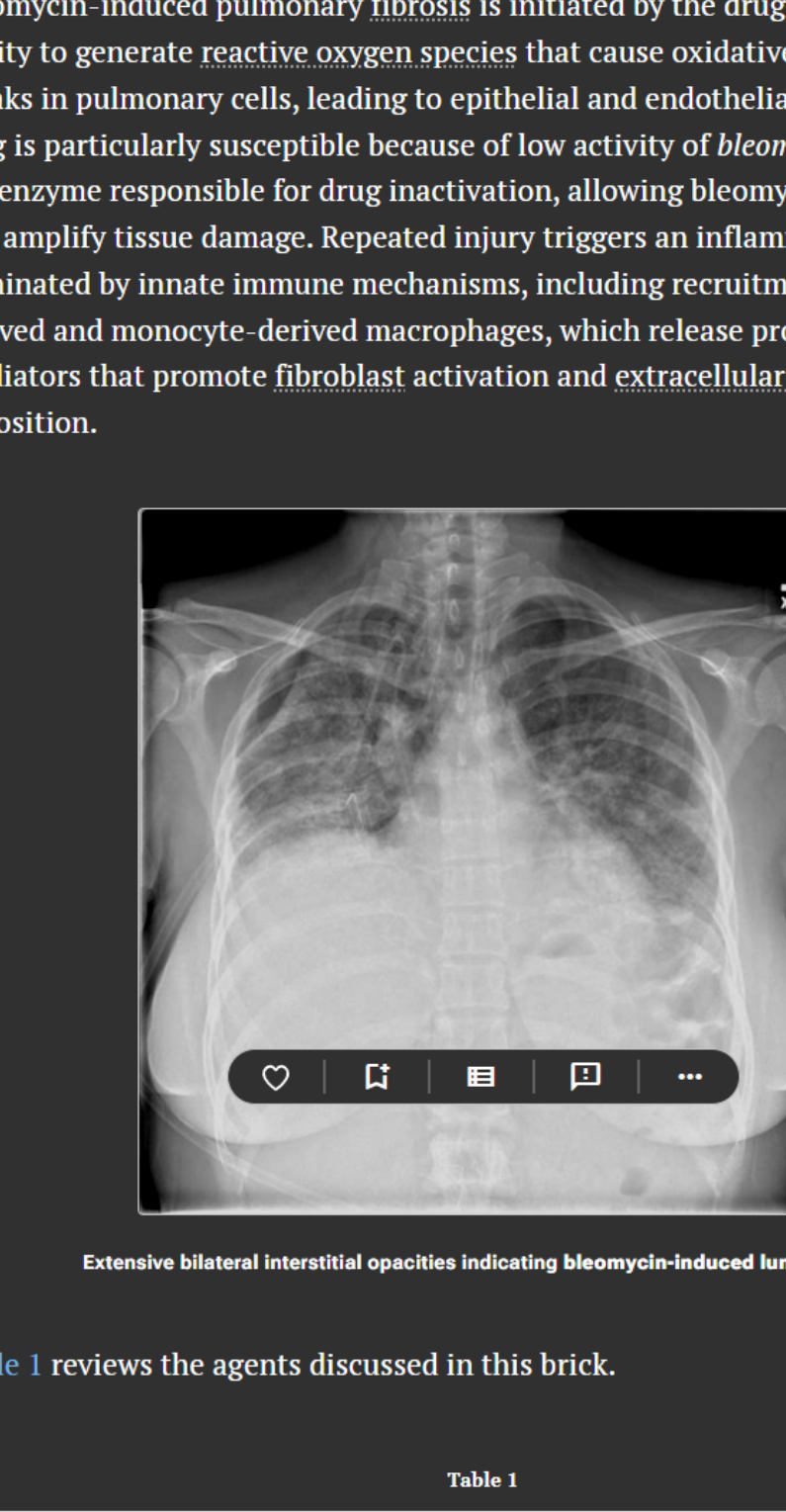
Bleomycin breaks DNA strands, probably by inducing free radical formation. It is cell-cycle specific in the G_2 and M phases. Bleomycin is a small peptide that contains an Fe-bindi molecule. DNA–bleomycin–Fe2+ complex appears to undergo oxidation to bleomycin–Fe3+. The free electrons react with oxygen to form superoxide or hydroxyl radicals, which, in turn, attack the phosphodiester bonds of DNA. The attack of the DNA bonds results in strand breakage and chromosomal aberrations. Pulmonary fibrosis is seen in 10% of patients.

Adverse effects of Bleomycin

Pulmonary toxicity is dose-limiting for bleomycin and usually presents as pneumonitis with cough, dyspnea, dry inspiratory crackles on physical examination, and infiltrates on chest x-ray. Pulmonary toxicity can occur in 6-10% of patients on bleomycin therapy. This is often referred to as "bleomycin lung". The incidence of pulmonary toxicity is increased in patients older than 70 years of age. Bleomycin is also unusual because myelosuppression is rare.



Bleomycin-induced pulmonary fibrosis is initiated by the drug's intrinsic ability to generate reactive oxygen species that cause oxidative DNA strand breaks in pulmonary cells, leading to epithelial and endothelial injury. The lung is particularly susceptible because of low activity of *bleomycin hydrolase*, the enzyme responsible for drug inactivation, allowing bleomycin to persist and amplify tissue damage. Repeated injury triggers an inflammatory response dominated by innate immune mechanisms, including recruitment of tissue-derived and monocyte-derived macrophages, which release profibrotic mediators that promote fibroblast activation and extracellular matrix deposition.



Extensive bilateral interstitial opacities indicating bleomycin-induced lung toxicity

Table 1 reviews the agents discussed in this brick.

Table 1

Drug Type	Examples	Action	Toxicity
Topoisomerase inhibitors	Topotecan, etoposide	Inhibit DNA ligation	Myelosuppression
Microtubule inhibitors			
	Vinca alkaloids	Block microtubule polymerization	Peripheral neuropathy
	Taxanes	Block microtubule degradation	Peripheral neuropathy
Antitumor antibiotics			
	Anthracyclines	Intercalate into DNA	Cardiac toxicity
	Mitomycin C	Alkylate DNA	Hemolytic uremic syndrome
	Bleomycin	Break DNA	Pulmonary fibrosis

CASE CONNECTION

BACK TO INTRODUCTION ↑

Thinking back at BV, est ordered for him?

You discussed with BV that the lung function test was ordered for him because he is on Bleomycin. Bleomycin is used to treat several types of cancers, including lymphomas, due to its broad activity and low myelotoxicity. You explain to BV that pulmonary toxicity can occur in 6-10% of the patients on bleomycin therapy. The toxicity can cause shortness of breath, cough, and can also lead to the development of pulmonary fibrosis. Pulmonary fibrosis can be fatal and non-reversible for the patient. Pulmonary function tests can help establish a baseline of lung function, and if signs of toxicity appear, appropriate measures can be taken to prevent irreversible damage. You again reassure BV that the oncology team will help him throughout this treatment.

Summary

- Topoisomerase inhibitors, including topotecan and irinotecan (topoisomerase I) and etoposide and teniposide (topoisomerase II), prevent DNA strand religation, leading to cell death.
- Microtubule inhibitors, such as vinca alkaloids and taxanes, disrupt mitotic spindle function, causing cell cycle arrest and apoptosis.
- Vinca alkaloids inhibit microtubule polymerization, while taxanes stabilize microtubules, preventing depolymerization.
- Topoisomerase inhibitors adverse effects commonly include myelosuppression, nausea, vomiting, and fatigue; irinotecan is associated with high rates of diarrhea, with early cholinergic symptoms treatable by atropine, and late diarrhea posing risks of dehydration, electrolyte imbalance, and sepsis
- Common adverse ude peripheral neuropathy, with vinca alkaloids also causing hair loss and taxanes leading to neutropenia.
- Anthracyclines, like doxorubicin, intercalate into DNA, causing apoptosis and cardiotoxicity, with dexrazoxane used to mitigate heart damage.
- Bleomycin induces DNA strand breaks, leading to pulmonary fibrosis, while mitomycin C acts as an alkylating agent, potentially causing hemolytic uremic syndrome.

Review Questions

1. Which of the following drug classes or subclasses acts by preventing the depolymerization of microtubules, thus inhibiting the progression of the cancer cell cycle from metaphase to anaphase?

A. ○ Mitomycin C
B. ● Taxanes
C. ○ Topoisomerase I inhibitors
D. ○ Topoisomerase II inhibitors
E. ○ Vinca alkaloids

Hide Explanation

The correct answer is ubulin and stabilizing the microtubular structuring, inhibiting the depolymerization of the microtubule and the progression of mitosis from metaphase to anaphase. Mitomycin C (A) is an antitumor antibiotic that alkylates DNA. Topoisomerase I inhibitors (C) and topoisomerase II inhibitors (D) bind to topoisomerase and prevent the religation of DNA strands during replication, transcription, and repair. Vinca alkaloids (E) are antimicrotubule drugs like the taxanes, but they act by preventing the polymerization of the microtubule.

2. Which phase do topoisomerase II inhibitors target?

A. ○ Late M phase
B. ● S or early G2 phase
C. ○ Early G1 phase
D. ○ Late G2 phasee
E. ○ G0 phase

Hide Explanation

The correct answer is S phase or early G2 phase (B). Topoisomerase II inhibitors like etoposide and teniposide act during these phases. Late M phase (A) is not targeted by these inhibitors. G1 phase (C) and late G2 phase (D) are also not the specific targets. The G0 phase (E) is a resting phase and not targeted by these drugs.

3. Which agent is used as a cardioprotective agent against Doxorubicin-induced toxicity?

A. ● Dexrazoxane
B. ○ Methylene blue
C. ○ Dexamethasone
D. ○ Mesna

Hide Explanation

The correct answer is Dexrazoxane (A). By chelating the iron-mediated free radicals, it limits tissue damage caused by anthracyclines such as doxorubicin. Answer choice (B), Methylene blue is used in methemoglobinemia and not in anthracycline toxicity. Choice (C), Dexamethasone is a steroid used as pretreatment. Answer (D), Mesna is used to reverse cyclophosphamide-induced nephrotoxicity.

4. Which of the following is most likely to cause diarrhea with cholinergic symptoms?

A. ○ Bleomycin
B. ○ Vincristine
C. ○ Paclitaxel
D. ● Irinotecan

Hide Explanation

The correct answer is Irinotecan (D). Irinotecan can cause acute and chronic diarrhea, which is also associated with cholinergic symptoms. The incidence of diarrhea can be as high as 80%. Answer (A), bleomycin is associated with pulmonary toxicity, while vincristine (B) and Paclitaxel (C) are associated with peripheral neuropathy and hypersensitivity reactions. These effects also occur independently of cholinergic symptoms.

How would you rate this Brick?

Review Learning Objectives

Check off the objectives when you feel comfortable with the concept

01 ☑ ~~Explain the mechanism of action of agents of topoisomerase inhibitors, microtubule inhibitors, and antitumor antibiotics.~~

02 ☑ ~~Explain the adverse effects of topoisomerase inhibitors, microt~~

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Objective Confidence: 2/2